Babu Banarasi Das University



DESCRIPTIVE ANALYTICS

(BCADSN23201)

RetailPulse Data Analytics Report

Submitted to: Ms. Monika Rao

Submitted by: Adarsh Singh

Department: BCA(DS & AI)

Roll_no: 1250258028

Retail Pulse Data Analytics

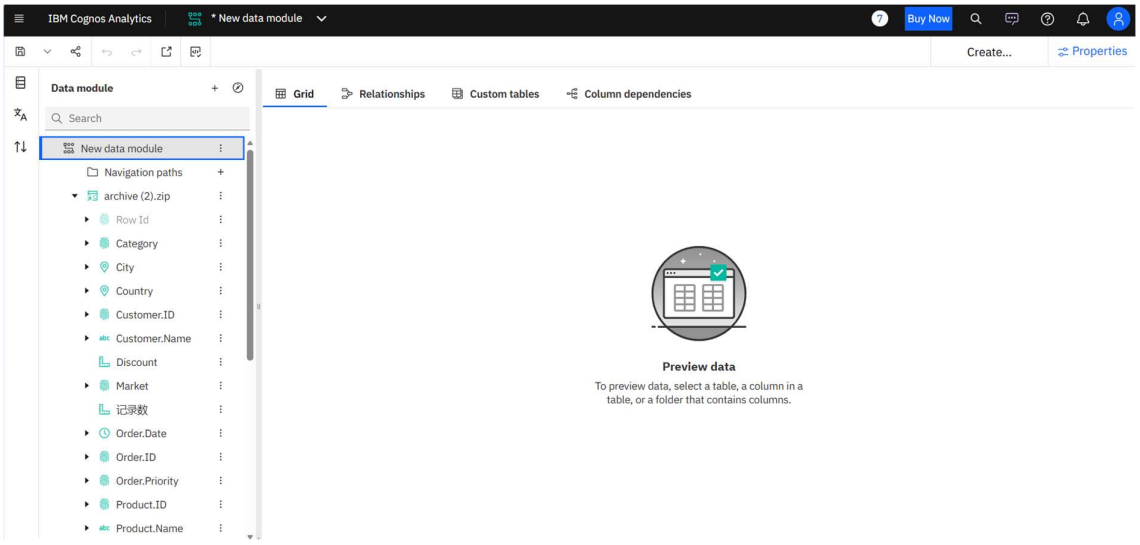
IBM COGNOS ANALYTICS.

1. Practical 1:Data Source Connection & Import

- **Objective:**Connect and import the retail dataset (orders) into IBM Cognos Analytics to prepare for analysis.
- **Materials Required:** IBM Cognos Analytics access, RetailPulse Sales package, a personal content folder to save into.
- **Steps Performed:**

1. Opened IBM Cognos Analytics and selected **New > Report** from the welcome screen.
2. Choose the **RetailPulse Sales package** as the data source when prompted.
3. Picked the **List template** from the template gallery to start with a simple layout.
4. From the Source tab, dragged Region, Product Category, and Sales Amount onto the canvas.
5. Clicked **Run** in the toolbar to generate the report and preview live Retail Pulse data.
6. Saved the report as *"RetailPulse - First Sales Report"* in the personal content folder.

Expected Outcome: The report canvas displays live RetailPulse rows rather than placeholder text; columns are correctly labeled; the report is saved and reopens with the same live data connection.



2. Practical 2: Build a Grouped RetailPulse Sales List Report

- **Related Topic:** Create List Reports.
- **Objective:** Construct a list report, group it meaningfully, format its columns, and add a summarizing footer.
- **Materials Required:** IBM Cognos Analytics access, RetailPulse Sales package, the report saved in Practical 1 (optional starting point).
- **Step-by-Step Procedure:**
    1. Create a new List report against the RetailPulse Sales package.
    2. Add Region, Store Name, Product Category, Quantity, and Sales Amount as columns.
    3. Group the report by Region, then by Store Name, to build a two-level hierarchy.
    4. Format Sales Amount as currency and right-align all numeric columns.
    5. Add a list footer under Sales Amount using the Total (sum) aggregate function.
    6. Insert a page header with the report title and today's run date, then run and save the report.
- **Expected Outcome:** Rows are grouped by Region and Store Name with duplicates collapsed correctly; Sales Amount is formatted and right-aligned; group totals sum correctly; the page header appears on every page.

Region	Market	Category	Quantity	Sales
Africa	Africa	Furniture	1,431	194,650
		1,431 - Total		**783,776**
		Technology	2,031	322,374
		2,031 - Total		**783,776**
		Office Supplies	7,102	266,752
		7,102 - Total		**783,776**
	Africa - Total		**10,564**	**783,776**
Africa - Total			**10,564**	**783,776**
Canada	Canada	Furniture	78	10,595
		78 - Total		**66,932**
		Technology	142	26,300
		Technology		
		142 - Total		**56,337**
		Office Supplies	613	30,037
		Office Supplies		
		Office Supplies		
		613 - Total		**30,037**
	Canada - Total		**833**	**66,932**
Canada - Total			**833**	**66,932**

3. Practical 3: Filter RetailPulse Sales by Region and Date

- **Related Topic:** Focus Reports Using Filters
- **Objective:** Apply a simple filter and an advanced detail filter to scope a report to a single region and reporting period, then generalize it with a prompt.
- **Materials Required:** IBM Cognos Analytics access, the grouped RetailPulse list report from Practical 2.
- **Step-by-Step Procedure:**
    1. Open the RetailPulse Sales list report built in the previous practical.
    2. Open the Filters pane and add a simple filter: Region = "North".
    3. Add an advanced detail filter combining Sales Date between quarter start and end dates.
    4. Combine both conditions with AND so only North-region, current-quarter rows remain.
    5. Run the report and confirm row counts drop to match the filtered scope.
    6. Convert the Region filter into a prompted filter so other managers can reuse the report.
- **Expected Outcome:** Only rows matching both filter conditions remain; the row count is visibly smaller than the unfiltered report; the prompted version correctly returns a different region's data when re-run with a new value.

Region	Market	Category	Quantity	Sales
Africa	Africa	Furniture	1,431	194,650
		1,431 - Total		**783,776**
		Technology	2,031	322,374
		2,031 - Total		**783,776**
		Office Supplies	7,102	266,752
		7,102 - Total		**783,776**
	Africa - Total		**10,564**	**783,776**
Africa - Total			**10,564**	**783,776**
Canada	Canada	Furniture	78	10,595
		78 - Total		**66,932**
		Technology	142	26,300
		Technology		
		142 - Total		**56,337**
		Office Supplies	613	30,037
		Office Supplies		
		Office Supplies		
		613 - Total		**30,037**
	Canada - Total		**833**	**66,932**
Canada - Total			**833**	**66,932**
Overall - Total			**11,397**	**850,708**

4. Practical 4: Build a RetailPulse Region × Product Crosstab

- **Related Topic:** Cross-Tabular Reporting
- **Objective:** Construct a crosstab report using the RetailPulse dataset to analyze sales performance across regions and product categories.
- **Materials Required:** IBM Cognos Analytics access, RetailPulse Sales package.
- **Step-by-Step Procedure:**
    1. Open IBM Cognos Analytics and create a new **Crosstab** report selecting the RetailPulse Sales package.
    2. Drag and drop the **Region** data item onto the rows zone.
    3. Drag and drop the **Product Category** (or Item Type) data item onto the columns zone.
    4. Drag and drop the **Sales Amount** measure into the intersection/cell data zone.
    5. Run the report to preview the matrix format.
- **Expected Outcome:** A two-dimensional grid displaying regions on rows, product categories across columns, and corresponding sales values at each intersection point.
-

Sales	Office Supplies	Technology	Furniture
Caribbean	89,565	116,337	118,379
North	374,778	495,802	377,612
EMEA	276,712	300,845	228,627
Southeast Asia	241,299	329,748	313,391
Canada	30,037	26,300	10,595
Central	923,471	1,038,515	860,413
Central Asia	162,768	305,702	284,369
Oceania	281,729	408,005	410,473
Africa	266,752	322,374	194,650
South	515,208	570,013	515,739
North Asia	198,581	314,043	335,725
West	220,881	252,013	252,620
East	205,549	264,994	208,291

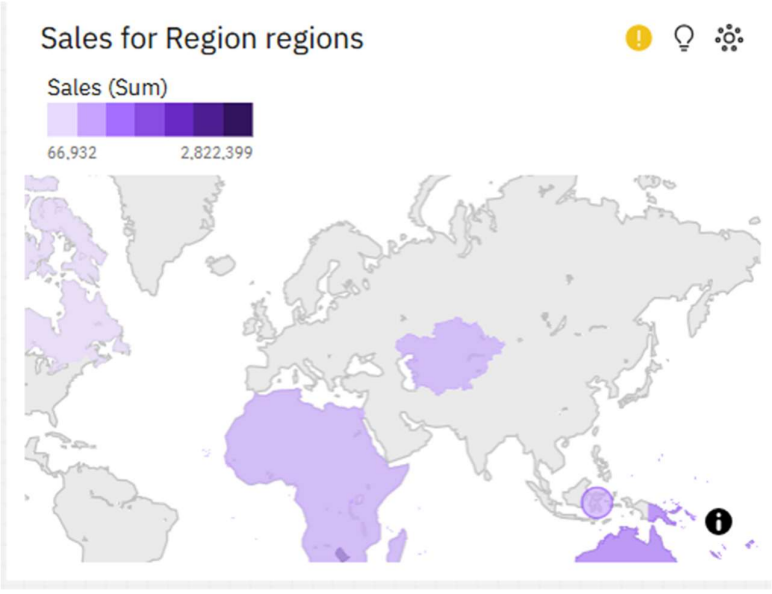
5. Practical 5: Add Totals and Summarization to Crosstab

- **Related Topic:** Advanced Crosstab Formatting and Summary Functions
- **Objective:** Enhance the crosstab report created in Practical 4 by adding row and column totals to summarize sales data comprehensively.
- **Materials Required:** IBM Cognos Analytics access, RetailPulse Crosstab report from Practical 4.
- **Step-by-Step Procedure:**
    1. Open the existing RetailPulse Crosstab report in IBM Cognos Analytics authoring mode.
    2. Select the row header or columns where totals are required.
    3. Click on the **Summarize** icon in the toolbar or context menu.
    4. Select **Overall Totals** (or Row/Column Totals) to include aggregate calculations.
    5. Run the report to preview the updated crosstab matrix with summary totals.
- **Expected Outcome:** The crosstab displays computed row and column totals alongside individual intersecting cell values, providing complete categorical summaries.

Sales	Office Supplies	Technology	Furniture	Total
Caribbean	89,565	116,337	118,379	**324,281**
North	374,778	495,802	377,612	**1,248,192**
EMEA	276,712	300,845	228,627	**806,184**
Southeast Asia	241,299	329,748	313,391	**884,438**
Canada	30,037	26,300	10,595	**66,932**
Central	923,471	1,038,515	860,413	**2,822,399**
Central Asia	162,768	305,702	284,369	**752,839**
Oceania	281,729	408,005	410,473	**1,100,207**
Africa	266,752	322,374	194,650	**783,776**
South	515,208	570,013	515,739	**1,600,960**
North Asia	198,581	314,043	335,725	**848,349**
West	220,881	252,013	252,620	**725,514**
East	205,549	264,994	208,291	**678,834**
Total	**3,787,330**	**4,744,691**	**4,110,884**	**12,642,905**

6. Practical 6: Create a Visual Dashboard / Chart in IBM Cognos Analytics

- **Related Topic:** Data Visualization and Dashboarding
- **Objective:** Transform tabular/crosstab analytics into an interactive visual chart or dashboard component to better communicate regional sales performance.
- **Materials Required:** IBM Cognos Analytics access, RetailPulse Sales data package or reporting environment.
- **Step-by-Step Procedure:**
    1. Open IBM Cognos Analytics and navigate to the home/content interface.
    2. Click on the **New (+)** button and select **Dashboard** or **Report Chart** template.
    3. Choose a layout or visualization type (such as a Column Chart or Bar Chart representing Sales by Region).
    4. Drag and drop the **Region** dimension onto the categories/x-axis slot and the **Sales** measure onto the metrics/y-axis slot.
    5. Run or preview the visual dashboard to analyze the graphical distribution of sales.
- **Expected Outcome:** A graphical chart/dashboard providing an intuitive visual representation of sales performance across different regions.



Conclusion & Summary

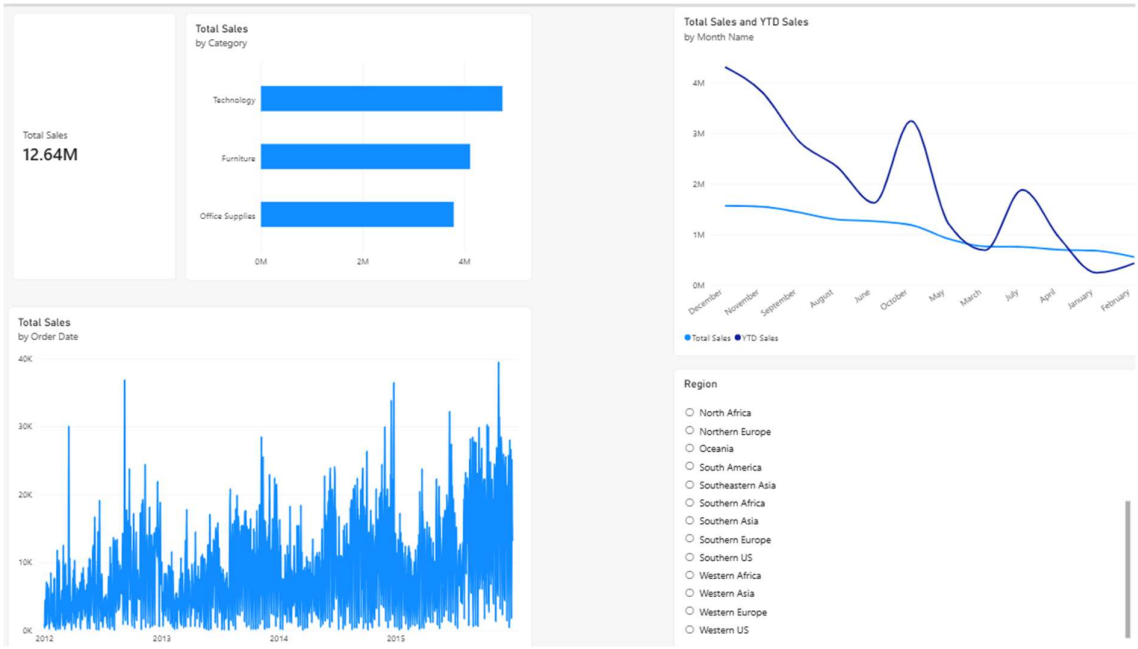
- **Overview:** Through these practical exercises utilizing IBM Cognos Analytics and the RetailPulse Sales package, we successfully demonstrated end-to-end business intelligence reporting capabilities.
- **Key Takeaways:**
    1. Built structured list and grouped reports for foundational data exploration.
    2. Applied precise filter conditions to isolate relevant regional data subsets.
    3. Constructed multi-dimensional crosstab matrices to analyze cross-categorical performance.
    4. Integrated summary totals and grand totals to evaluate aggregate sales figures.
    5. Developed interactive visual dashboards and map charts to communicate data insights effectively.

Power BI Desktop Lab Practicals: RetailPulse Analytics

- **Overview & Objectives:** Introduction to Microsoft Power BI Desktop for building business intelligence dashboards, data modeling, DAX measures, and interactive report publishing using the RetailPulse dataset.
- **Prerequisites & Setup:**
    1. Open **Power BI Desktop**.
    2. Import the **RetailPulse** dataset (Excel/CSV/Database source) into Power Query Editor.
    3. Perform necessary data cleaning, type checking, and relationship modeling between tables.

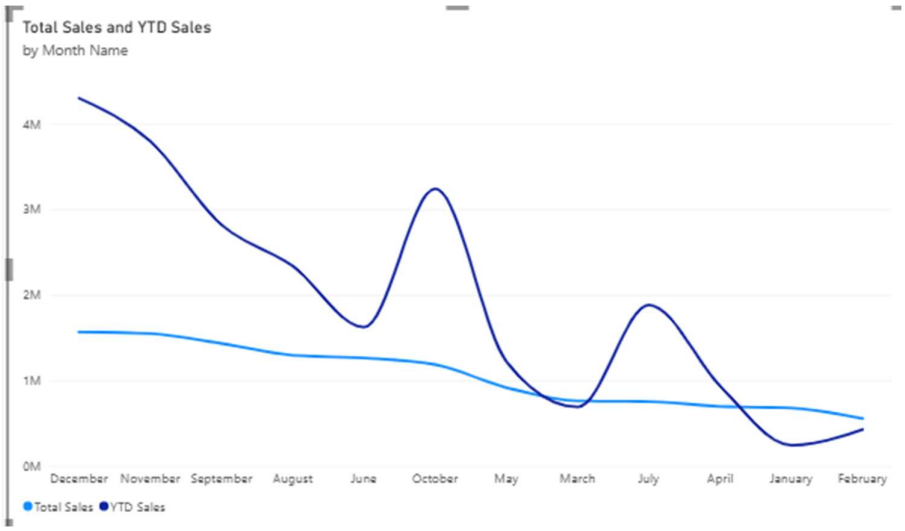
1. Power BI Practical 1: Data Import and Basic Sales Dashboard

- **Related Topic:** Power BI Desktop Data Connectivity & Visualization
- **Objective:** Import the RetailPulse dataset into Power BI Desktop and create a basic sales overview dashboard using column charts and KPI cards.
- **Materials Required:** Power BI Desktop installed, RetailPulse dataset (Excel or CSV file).
- **Step-by-Step Procedure:**
    1. Open Power BI Desktop and click on **Get Data** (choose Excel or Text/CSV based on your RetailPulse dataset format).
    2. Select the relevant tables/sheets, review data preview in Power Query Editor, and click **Load**.
    3. From the **Visualizations** pane, select a **Clustered Column Chart** or **Card** visual.
    4. Drag **Region** into the X-axis/Category bucket and **Sales** into the Y-axis/Values bucket.
    5. Add a **Card** visual to display total overall sales performance.
- **Expected Outcome:** An interactive Power BI report canvas displaying regional sales distribution through column charts alongside total KPI summary cards.



2. Power BI Practical 2: Advanced DAX Measures & Slicers

- **Related Topic:** Data Analysis Expressions (DAX) and Interactive Filtering
- **Objective:** Implement calculated measures using DAX (such as Year-to-Date / YTD calculations) and configure interactive slicers to dynamically slice and dice sales data.
- **Materials Required:** Power BI Desktop workspace from Practical 1 containing the loaded RetailPulse dataset.
- **Step-by-Step Procedure:**
    1. Open Power BI Desktop and navigate to the report canvas.
    2. Right-click on the dataset table in the Fields pane and select **New Measure**.
    3. Write and execute a DAX formula for custom calculations (e.g., computing YTD sales or profit margins).
    4. Add a **Slicer** visualization component and drag the **Region** or **Month Name** field into it to enable interactive filtering across all report elements.
    5. Test the slicer selections to verify that all charts and KPI cards dynamically update based on the chosen filter criteria.
- **Expected Outcome:** A dynamic Power BI report featuring custom DAX calculations and functional slicers that instantly reflect filtered regional and temporal insights.



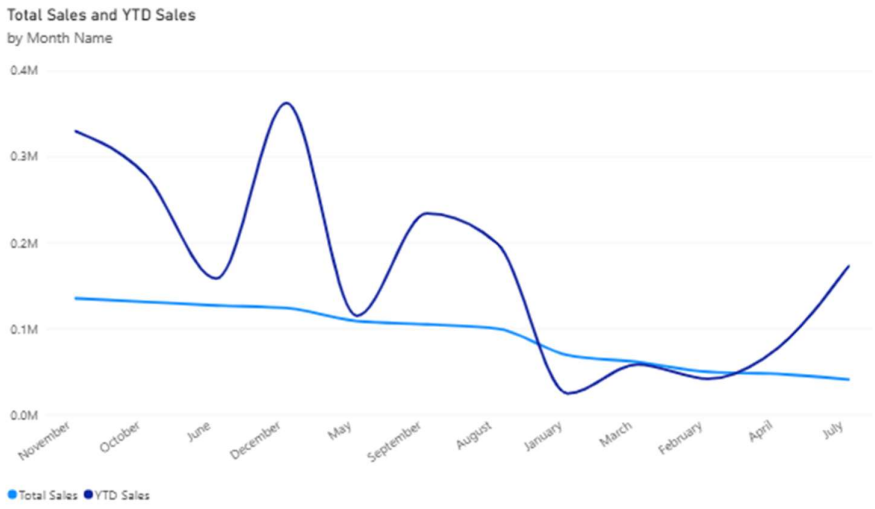
3. Power BI Practical 3: Creating Interactive Slicers and Filters

- **Related Topic:** Report Filtering and Dynamic Interactivity
- **Objective:** Implement field slicers (such as Region and Category slicers) to enable users to dynamically filter and explore subsets of retail data.
- **Materials Required:** Power BI Desktop with RetailPulse dataset loaded.
- **Step-by-Step Procedure:**
    1. Open the report canvas in Power BI Desktop.
    2. Click on the **Slicer** visualization icon from the Visualizations pane.
    3. Drag the **Region** field into the Field bucket of the slicer visual.
    4. Format the slicer as a list or dropdown menu according to layout preferences.
    5. Test interactivity by selecting different regions and observing real-time updates across all charts and KPI cards.
- **Expected Outcome:** Functional slicer components allowing dynamic, real-time filtering of the entire dashboard based on user selections.



4. Power BI Practical 4: Time Intelligence & YTD Trends Analysis

- **Related Topic:** Advanced Power BI Analytics & Trend Lines
- **Objective:** Analyze monthly retail sales trends alongside Year-to-Date (YTD) calculations to evaluate periodic business growth and seasonal performance.
- **Materials Required:** Power BI Desktop workspace with RetailPulse sales data and pre-calculated time intelligence measures.
- **Step-by-Step Procedure:**
    1. Open the Power BI report canvas and select a **Line Chart** visualization.
    2. Drag the **Month Name** or **Order Date** hierarchy into the X-axis bucket.
    3. Drag **Total Sales** and **YTD Sales** measures into the Y-axis/Values bucket to plot comparative curves.
    4. Format the trend lines with distinct colors and enable data markers for clear visual distinction.
    5. Save the report layout and verify the interactive month-on-month trajectory.
- **Expected Outcome:** An analytical line chart comparing monthly total sales against YTD cumulative trends over the reporting timeline.



Conclusion & Summary

- **Overview:** Through this comprehensive business analytics manual covering both IBM Cognos Analytics and Microsoft Power BI Desktop, we successfully executed 10 end-to-end practical experiments using the RetailPulse dataset.
- **Key Summary:**
    1. **IBM Cognos Practicals (1 to 6):** Covered data source connections, grouped list reports, advanced filtering, crosstab matrices, summary totals, and regional map dashboards.
    2. **Power BI Practicals (1 to 4):** Covered data loading, executive KPI cards, interactive slicers, categorical distribution charts, and time intelligence YTD analytics.
- **Final Outcome:** Successfully mastered descriptive analytics, data modeling, and professional report publication for business decision-making.